

SIRCA-RAG

Semi-Autonomous RAG for Peruvian Medicinal Plant Knowledge Integration

Technical Report — SimBig / WAIMLAp 2026

1. Architecture Overview

The SIRCA-RAG pipeline implements a Corrective RAG (CRAG) architecture with 4 stages:

- 1. Query Classification** — Rule-based classifier (factual / exploratory / comparative) that adjusts hybrid retrieval weights dynamically.
- 2. Hybrid Retrieval** — BGE-M3 dense embeddings (1024d) + BM25 sparse index, fused via Reciprocal Rank Fusion (RRF), reranked with cross-encoder (ms-marco-MiniLM-L-12-v2).
- 3. CRAG Evaluation** — Threshold-based decision: accept (≥ 0.35), refine (0.15–0.35), or web_search (< 0.15). Uses pre-computed rerank scores for sub-millisecond evaluation.
- 4. Grounded Generation** — Chain-of-Thought protocol (Extract → Verify → Compose) with strict grounding rules. Supports DeepSeek V4 Flash (API), Ollama Qwen3.5 (local), and template backends.

Query → Classify → Retrieve → Evaluate → [accept | refine | web_search] → Generate

2. Data Corpus

3,040 vectorized chunks from 8 Peruvian medicinal plant species.

Sources: PubMed, CrossRef, PeruNPDB, GBIF, WFO, SciELO

Embedding model: BAAI/bge-m3 (1024 dimensions)

Chunk size: 512 tokens, overlap: 64

#	Species	Common Name
1	Uncaria tomentosa	Cat's Claw / Una de Gato
2	Lepidium meyenii	Maca
3	Croton lechleri	Dragon's Blood / Sangre de Grado
4	Minthostachys mollis	Muna
5	Erythroxylum coca	Coca
6	Smallanthus sonchifolius	Yacon
7	Physalis peruviana	Aguaymanto / Cape Gooseberry
8	Buddleja incana	Quishuar / Kiswar

3. Evaluation Results (DeepSeek Backend)

Metric	Score	Target	Status
BERTScore F1 (roberta-large)	0.9028	≥ 0.90	PASS
Semantic Similarity (cross-encoder)	0.9929	—	—
Context Precision	1.0000	—	—
Context Recall	1.0000	—	—
MRR	1.0000	—	—
NDCG@5	1.0000	—	—
Entity Recall	0.7946	—	—
Faithfulness	0.8411	≥ 0.80	PASS
Answer Relevancy	0.9995	—	—

4. Key Technical Decisions

Chain-of-Thought Grounding: 3-step protocol prevents data invention. The model internally (1) extracts facts from context, (2) verifies each against sources, (3) composes the final answer. Only Step 3 is shown to the user.

Hybrid Faithfulness Metric: 65% semantic (cross-encoder sentence-level scoring via ms-marco-MiniLM-L-12-v2 with sigmoid normalization) + 35% lexical (content-word overlap with bilingual stopword filtering EN+ES).

BERTScore with roberta-large: DeBERTa variants caused OverflowError on this system. roberta-large provides stable $F1 \geq 0.90$.

Temperature 0.0: Deterministic generation for reproducible evaluation across runs.

Bilingual Support: System prompt instructs model to answer in the same language as the query (Spanish or English). Boilerplate filtering handles both languages.

5. Pipeline Trace Example

Query: "What are the main alkaloids in Uncaria tomentosa?"

Node	Duration	Details
classify	4ms	category: exploratory, confidence: 0.60
retrieve	3,195ms	5 documents, hybrid alpha: 0.6, BGE-M3 + BM25 + rerank
evaluate	<1ms	action: accept, confidence: 0.89 (pre-computed scores)
generate	varies	DeepSeek: ~30-150s Template: <1ms Ollama: ~10-30s

6. Web Service API

Method	Path	Description
GET	/	Interactive dark-themed frontend with example queries
GET	/api/health	System status, backend availability, vectorstore size
POST	/api/query	Full pipeline: query, backend, language & rarr; answer, citations, trace
GET	/api/species	List of 8 target species

7. Deployment (Docker / Dokploy)

Image: python:3.11-slim

Port: 8000

Environment variables: DEEPSEEK_API_KEY, OLLAMA_BASE_URL

Health check: GET /api/health (30s interval, 60s start period)

```
docker compose up --build
```